

42.510 Statistical Learning for Data Science

Statistics review

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Week 1, Lecture 2

Probability vs. Statistics

- *Probability* lets us work with models that we can use to generate random variables, quantities and simulated data. For example, we can obtain i.i.d samples from our model. Probability creates a framework for us to model randomness.
- *Statistics* is based on the real world. We work with data sets that we do not know much of, we come up with a set of assumptions about the data set which allows us to return to the probability models so that we can say something concrete about the data set.
- *Learning*: using data to understand real world phenomenon, may or may not always be based on probabilistic assumptions.

How we work with data sets in statistics?

- 1 Assume they are i.i.d observations.
 - 2 Assume they come from a probability distribution F which belongs to a collection of all distributions.
 - 3 We want to infer about F .
- Note that this is *learning* as well, but in statistics we call it *inference*.

Statistics: notations and terminology

- *Random variables:* X, Y, Z .
- *Realizations, observations, observed values:* x, y, z .
- *Data:* A collection of realization of random variables.
- *Sample:* Random sample of size n : $X_1, \dots, X_n$ iid from some distribution F .
- *Realization of a random sample:* $x_1, \dots, x_n$.
- *Statistic:* a function of the data.
- *Parameter:* true value that numerically characterizes a population value. This is often not possible to know exactly in practice. A statistic is often used to estimate such parameters.
- *Sample mean:* $\bar{X}_n$ where $x_1, \dots, x_n$ are realizations of $X_1, \dots, X_n$.
- We may call $\bar{X}_n$ to be sample mean as well.

Descriptive statistics formulas

- Let $(X, Y) \sim F$ with $X \sim F_1, Y \sim F_2$.
- Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be iid from F .
- Let $(x_1, y_1), \dots, (x_n, y_n)$ be realizations from the above sample.

Name	Parameter	Statistic (realization)
Mean	$\mu_X = \mathbb{E}(X)$	$\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i$
Variance	$\sigma_X^2 = \text{Var}(X)$	$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)^2$
Std. dev.	$\sigma_X = \sqrt{\text{Var}(X)}$	$s_x = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)^2}$
Covariance	$\sigma_{XY} = \mathbb{E}(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))$	$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y})$
Correlation	$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$	$r_{xy} = \frac{s_{xy}}{s_x s_y}$

Estimating the number of German tanks

- During World War II, the Allied forces had noted the serial numbers of some captured/destroyed German tanks and wanted to estimate the total number of German tanks deployed.
- They correctly guessed that the serial numbers ran from 1 to N , where N was the total number of tanks produced.
- **QUESTION:** How can one estimate N ?

Month of production	Intelligence estimate	Statistical estimate	Actual (from post-war captured documents)
June 1940	1000	169	122
June 1941	1550	244	271
August 1942	1550	327	342

Estimating the number of German tanks

- **Goal:** Estimate the value of N from a sample of discrete uniform random variable taking values in $\{0, 1, \dots, N\}$. We are given a sample of size n drawn without replacement.
- Let the sample observations be $\{x_1, \dots, x_n\}$.
- Order the sample: $x_{(1)} \leq \dots \leq x_{(n)}$.
- Some possible estimates for N are:
 - 1 $\bar{x}_n + 3s_x$
 - 2 $2\bar{x}_n - 1$
 - 3 $x_{(n)}$
 - 4 $x_{(n)} + x_{(1)} - 1$
 - 5 $\frac{n+1}{n}x_{(n)} - 1$
- Estimate N if the numbers you got are:
 - ▶ 588; 601; 126; 699; 464; 468; 258; 440; 275; 320
- **R exercise:** Draw 500 simulated samples, each of size $n = 5$, from a population with $N = 250$. Evaluate the estimators. Compare variability, bias.

Simulation: R code

```
T<- 250 ## actual number of tanks
k<- 5 ## number of tanks seen
n<- 500 ## number of simulations

## n simulations of sample size k from 1,...,T
tanks<-matrix(replicate(n,sample(1:T,k)),nrow=n,ncol=k,byrow = "TRUE")

## Initialize matrices for storing the estimates
Ests<-matrix(rep(0,n*k),nrow=n,ncol=k)

Ests[,1]<- apply(tanks,1,function(x) mean(x)+3*sqrt(var(x))) ## Estimator 1
Ests[,2]<- apply(tanks,1,function(x) 2*mean(x)-1)           ## Estimator 2
Ests[,3]<- apply(tanks,1,function(x) max(x))                 ## Estimator 3
Ests[,4]<- apply(tanks,1,function(x) max(x)+min(x)-1)        ## Estimator 4
Ests[,5]<- apply(tanks,1,function(x) (k+1)*max(x)/k-1)        ## Estimator 5

mean_ests<-apply(Ests,2, function(x) mean(x))                ## Mean of the estimates

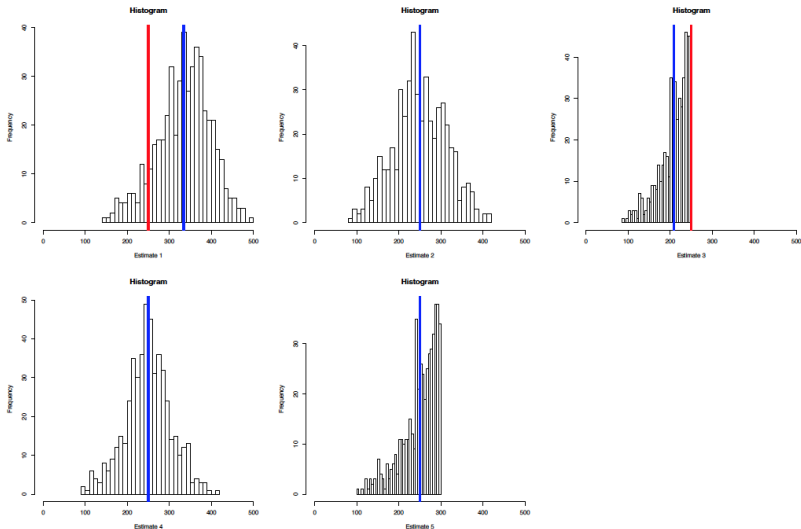
sd_ests<-apply(Ests,2,function(x) sd(x))                     ## Sd of the estimates

par(mfrow=c(2,3))
for(i in 1:5)
{hist(Ests[,i],density=FALSE,freq=T,breaks = 30, main="Histogram",xlab=paste("Estimate",i),xlim=c(0,500))
abline(v=250,col="red",lwd=3)
abline(v=mean_ests[i],col="blue",lwd=3)
}

dft<-data.frame(est=c("Est1","Est2","Est3","Est4","Est5"), mean=mean_ests,sd=sd_ests)

print(dft,digits=5)
```

Estimating the number of German tanks



- Histogram from a run

Estimating the number of German tanks

	est	mean	sd
1	Est1	334.21	64.06
2	Est2	250.06	63.54
3	Est3	209.10	34.34
4	Est4	249.82	54.78
5	Est5	249.92	41.21

- Estimates from the run.
- **Exercise:** Theoretically analyze the five different estimators with respect to variance, bias and compare.

Estimating the number of German tanks

- Estimate N if the numbers you got are:
 - ▶ 588; 601; 126; 699; 464; 468; 258; 440; 275; 320
- Some possible estimates for N are:
 - 1 $\bar{x} + 3s$ ($\hat{N}_1 = 960$)
 - 2 $2\bar{x} - 1$ ($\hat{N}_2 = 847$)
 - 3 $x_{(n)}$ ($\hat{N}_3 = 699$)
 - 4 $x_{(n)} + x_{(1)} - 1$ ($\hat{N}_4 = 824$)
 - 5 $\frac{n+1}{n}x_{(n)} - 1$ ($\hat{N}_5 = 768$)
- I chose $N = 735$.

Key concepts in Statistical inference

- Given observations/ realizations $x_1, \dots, x_n$: data, assume that they are i.i.d. realizations from $X_1, \dots, X_n$ from some distribution F .

Questions

- Can we know F ? What can we say about F ?
 - Parametric approach: assume Normal, exponential, binomial and estimate the parameters.
 - Here again we may have frequentist and Bayesian approaches.
 - Non-parametric approach: no specific parametric model, estimate from the entire data
- Can we infer about specific functionals of F : mean, median, variance, skewness, etc.

What do we do in statistical inference?

- Point estimation
- Interval estimation
- Hypothesis testing

Point estimation

- Point estimate: single best estimate for a parameter, c.d.f, regression function, etc.
- Estimate vs Estimator
 - ▶ Estimate
 - ★ Point estimate of θ is denoted by $\hat{\theta}_n$ or $\hat{\theta}$.
 - ★ $\hat{\theta}$ is obtained from the observations (varies when you change the dataset).
 - ★ Of course, θ is a fixed unknown parameter.
 - ▶ Estimator
 - ★ Point estimator is a function of the random sample: $X_1, \dots, X_n$.
 - ★ $\hat{\theta}_n = g(X_1, \dots, X_n)$. This is a random variable.
- Methods of estimation (parametric)
 - ▶ Maximum likelihood estimation (MLE).
 - ▶ Method of moments: *match theoretical moments with empirical moments.*

Bias and variance trade-off

- Let $\hat{\theta}_n$ be an estimator for θ . We want to assess the quality of the estimator. (keep in mind our example)
 - ➊ $\text{Bias}(\hat{\theta}_n) = \mathbb{E}(\hat{\theta}_n) - \theta$.
 - An estimator $\hat{\theta}_n$ is unbiased if $\text{Bias}(\hat{\theta}_n) = 0$.
 - ➋ Standard error of $\hat{\theta}_n$ (variability of the estimator):
 - $\text{se}(\hat{\theta}_n) = \sqrt{\text{Var}(\hat{\theta}_n)} = \sqrt{\mathbb{E}(\hat{\theta}_n - \mathbb{E}(\hat{\theta}_n))^2}$
 - ➌ Mean Squared Error (MSE)
 - $\text{MSE}(\hat{\theta}_n) = \mathbb{E}(\hat{\theta}_n - \theta)^2$.

Theorem

$$\text{MSE}(\hat{\theta}_n) = (\text{Bias}(\hat{\theta}_n))^2 + \text{Var}(\hat{\theta}_n).$$

Consistency of point estimators

- A point estimator $\hat{\theta}_n$ is consistent for θ if as $n \rightarrow \infty$, either
 - ▶ $\hat{\theta}_n \xrightarrow{P} \theta$ (weak consistency)
 - ▶ $\hat{\theta}_n \xrightarrow{a.s.} \theta$ (strong consistency)

Theorem

If $\text{Bias}(\hat{\theta}_n) \rightarrow 0$ and $\text{se}(\hat{\theta}_n) \rightarrow 0$ as $n \rightarrow \infty$ then $\hat{\theta}_n \xrightarrow{P} \theta$.

- We can check this for the previous example (and later ones).

Maximum Likelihood Estimation (MLE)

- Introduced by R.A. Fisher in the 1920s.
- The idea is to pick the value of the parameter which maximizes the *probability* of observing our data.
- For example, suppose we want to estimate the parameter p of a Bernoulli distribution. A random sample of size 5 drawn from this distribution reads:

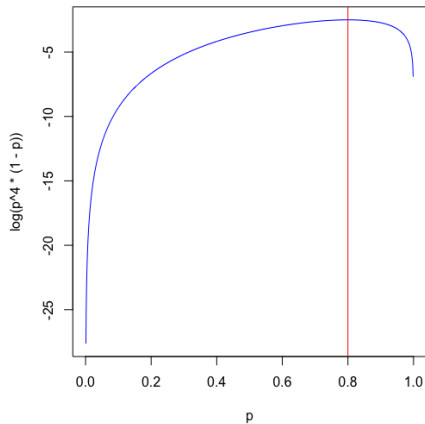
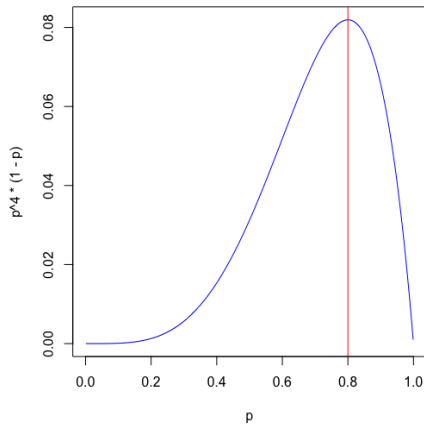
1, 1, 1, 0, 1.

- Which of these is most likely?

$$p = 0; p = 1/2; p = 2/3; p = 4/5; p = 1.$$

- Probability of this event = $p^4(1 - p)$.

MLE: Likelihood and log-likelihood



Likelihood function

- Given a probability distribution, let θ be a parameter to be estimated, and denote the probability density (or mass) function by $f(x|\theta)$.
- The **likelihood function** of θ , called $L(\theta)$ is the joint pdf (density) for a random sample of size n with observations $x_1, x_2, \dots, x_n$, is

$$L(\theta) := f(x_1, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta).$$

- If the maximum of this function occurs at $\hat{\theta}$, then $\hat{\theta}$ is the value of the parameter which maximizes the probability of observing our data, and we will use it as our estimate for θ .

Maximum likelihood estimate

- In many cases, we can find $\hat{\theta}$ by solving for

$$\frac{d}{d\theta} L(\theta) = 0, \quad \text{or equivalently,} \quad \frac{d}{d\theta} \ln(L(\theta)) = 0.$$

- We can use the *2nd derivative* to check that it is a maximum.
- Example: let $x_1, \dots, x_n$ be a random sample from Bernoulli(p). Then

$$L(\theta) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^{n\bar{x}} (1-p)^{n-n\bar{x}}.$$

- ▶ Solving for $\frac{d}{d\theta} \ln(L(\theta)) = 0$, we find that $\hat{p} = \bar{x}$.
- **Exercise:** Find the MLEs for μ and σ^2 where $x_1, \dots, x_n$ are observations from a normal distribution with parameters μ and σ^2 . The normal pdf is:

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

Are the estimators unbiased?

Adult persistence in head-turning asymmetry

- A neonatal right-side preference makes surprising romantic reappearance later in life.
- p = proportion of couples turning head to the right while kissing.
- Device an experiment to test if there is any truth in the statement.



Figure 2 The Kiss: the couple in Auguste Rodin's masterpiece are turning their heads to the right to kiss each other.

Human behaviour: Adult persistence of head-turning asymmetry, *Nature*, 2003. [Nature article](#)

Adult persistence in head-turning asymmetry

- Observe $n = 124$ couples while kissing.
- Collect data: 1 = right, 0 = left.
- Observed in study:
 - ▶ 80 out of 124 couples kissed from right.
- Estimate

$$\hat{p} = \frac{80}{124} = 64.5\%.$$

- Is there a preference to the right?
- Suppose $n = 3$ and you observe *right, right, left*. Is there a preference to the right?

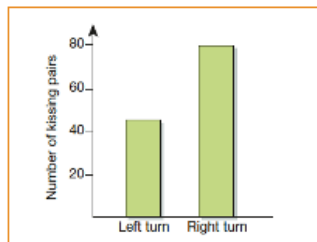


Figure 1 The number of couples who turn their heads to the right rather than to the left when kissing predominates by almost 2:1 (64.5%: 35.5%; $n = 124$ couples).

Adult persistence in head-turning asymmetry

Modeling

- $n = 124$.
- Define $X_i = \begin{cases} 1 & \text{if couple kisses from right,} \\ 0 & \text{otherwise.} \end{cases}$

Assumptions

- X_i 's are random variables.
- X_i 's are Bernoulli(p).
- X_i 's are independent.

Consequence

- **WLLN**: $\hat{p} = \bar{X}_n \xrightarrow{P} p$, hence p is close to $\hat{p} = 0.645$. (point estimation)
- But how close? Can we be confident that $p \neq 0.5$?

Adult persistence in head-turning asymmetry

Central limit theorem: confidence intervals (interval estimation)

- Using **Central limit theorem (CLT)** we have as $n \rightarrow \infty$:

$$\frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{p(1-p)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

- Hence for large n :

$$\bar{X}_n \approx \mathcal{N}(p, p(1-p)/n).$$

- If $Z \sim \mathcal{N}(0, 1)$ and we define z_α such that $\mathbb{P}(Z \leq z_\alpha) = 1 - \alpha$, then

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{p(1-p)}} \leq z_{\alpha/2}\right) \approx 1 - \alpha.$$

$$\text{or, } \mathbb{P}\left(\bar{X}_n - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \leq p \leq \bar{X}_n + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right) = 1 - \alpha.$$

Adult persistence in head-turning asymmetry

Central limit theorem: confidence interval (interval estimation)

- We know $p(1-p) \leq \frac{1}{4}$. Hence with $(1-\alpha)100\%$ confidence we can say that:

$$p \in \left[\bar{X}_n - z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}, \bar{X}_n + z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \right] \subseteq \left[\bar{X}_n - \frac{z_{\alpha/2}}{2\sqrt{n}}, \bar{X}_n + \frac{z_{\alpha/2}}{2\sqrt{n}} \right]$$

- Consider $\alpha = 0.05$, then $z_{0.025} = 1.96$. Hence with at least 95% confidence, we have approximately $p \in [0.56, 0.73]$.
- What happens when $n = 3$?
- We can't use CLT, but even with CLT, with 95% confidence we can say that $p \in [0, 1]$.
- Check *Hoeffding bounds* and use them to get a more useful result here.

Adult persistence in head-turning asymmetry

Central limit theorem: hypothesis testing

- We can also conduct a **hypothesis test** here.
- Suppose we actually had $p = \frac{1}{2}$. Then **CLT** will tell us

$$\mathbb{P} \left(\frac{\sqrt{n} \left| \bar{X}_n - \frac{1}{2} \right|}{\sqrt{\frac{1}{2} \times \frac{1}{2}}} \leq z_{\alpha/2} \right) \approx 1 - \alpha, \quad \text{or, } \mathbb{P} \left(2\sqrt{n} \left| \bar{X}_n - \frac{1}{2} \right| > z_{\alpha/2} \right) \approx \alpha$$

- So reject the premise that “ $p = \frac{1}{2}$ ” if $\left| \bar{X}_n - \frac{1}{2} \right|$ is a very large number.
- In statistical terminology, at level $\alpha = 0.05$, reject “ $p = \frac{1}{2}$ ” if $\left| \bar{X}_n - \frac{1}{2} \right| > \frac{z_{\alpha/2}}{2\sqrt{n}}$.
- With $\alpha = 0.05$, here we have $z_{0.025} = 1.96$, $n = 124$ and hence

$$\left| \bar{X}_n - \frac{1}{2} \right| = |0.645 - 0.5| = 0.145 \gg 0.09 = \frac{1.96}{2\sqrt{124}}.$$

- Hence we reject the premise or *hypothesis that* “ $p = \frac{1}{2}$ ” at level 0.05.

Hypothesis testing for means: chart

Purpose	Assumption on data	Test used	Formula	Distribution of statistic under H_0	Notations
Testing if sample comes from a population with a fixed hypothesized mean value	- Normal with known variance Or, the sample size n is large for CLT to hold ($n > 30$)	One sample z-test	$z = \frac{(\bar{x} - \mu)}{\frac{\sigma}{\sqrt{n}}}$	Normal	$\bar{x}$ is sample mean μ is population average σ is population s.d. s is sample s.d. (sometimes for large samples s replace σ for the z-test when σ is unknown) n is sample size
	Normal with unknown variance (small sample)	One sample t-test	$t = \frac{(\bar{x} - \mu)}{\frac{s}{\sqrt{n}}}$	t_{n-1}	
Testing for equality (or a particular value of difference) of mean between Population 1 and Population 2.	Both populations are Normal with known and not necessarily equal variance AND, both sample sizes are large for CLT to hold ($n_1 > 30, n_2 > 30$)	Two sample z-test	$z = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	Normal	$\bar{x}_1, \bar{x}_2$ are sample means μ_1, μ_2 are population averages σ_1, σ_2 are population s.d., may be replaced by s_1, s_2 for large samples when σ_1, σ_2 are unknown n_1, n_2 are sample sizes
	Both populations are Normal with unknown and not necessarily equal variance AND, both sample sizes are small.	Welch's (unequal variance) t-test	$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	t_v	The statistic follows an approximate t-distribution with d.f: $v \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\left(\frac{s_1^4}{n_1^2(n_1-1)} + \frac{s_2^4}{n_2^2(n_2-1)} \right)}$
	Both populations are Normal with unknown and equal variance AND, both sample sizes are small.	Two sample t-test	$\frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{s_p^2 * \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$	$t_{n_1+n_2-2}$	In addition to above $S_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ S_p^2 is their pooled variance or estimator of their common sd
	The populations are paired or dependent (and naturally with same sample size).	Paired t-test	$t = \frac{\bar{d}}{\left(\frac{s}{\sqrt{n}} \right)}$	t_{n-1}	$\checkmark$ d is average of the difference between the elements of the two samples $\checkmark$ s is the std. dev. of the difference between the elements of the two samples: $s = \sqrt{\frac{\sum_{i=1}^n d_i^2 - \frac{(\sum d_i)^2}{n}}{(n-1)}}$

Testing and distribution families

These are for you to explore.

- t -distribution and t -test (vs z -test).
- χ^2 -distribution and χ^2 -test of independence (non-parametric).
- F -distribution and F -test.

Student's t -distribution

- Defined on the entire real line like Normal distribution.
- Looks like normal (but heavier-tailed).
- Has one parameter which is positive integer-valued, which is called the **degree of freedom**.

Snedecor's F -distribution

- Defined on the non-negative reals.
- Has two parameters: they are positive integer-valued and also called degrees of freedom (d.f.).

References

- *Class notes* for Statistical learning for Data Science (2025).
- Wasserman (2004) *All of Statistics: a concise course in statistical inference* (Vol. 26).